
M1 Internship Report

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Abstract

The abstract paragraph should be indented ½ inch (3 picas) on both the left- and right-hand margins. Use 10 point type, with a vertical spacing (leading) of 11 points. The word **Abstract** must be centered, bold, and in point size 12. Two line spaces precede the abstract. The abstract must be limited to one paragraph.

1 Training and growing as a hybrid switched dynamic system

1.1 Training and growing as a hybrid switched dynamic system

Consider a hybrid system, that is, a system that exhibits dynamics with features that are typical of continuous-time systems and discrete systems.

A hybrid system $\mathcal{H} = (C, F, D, G)$ defined over $\mathbb{R}^n$ is specified by a flow set $C \subseteq \mathbb{R}^n$ and a flow dynamics $\dot{x} \in F(x)$ valid on C ; and by a jump set $D \subseteq \mathbb{R}^n$ and a jump map $x^+ \in G(x)$ valid on D .

The solutions evolve continuously according to F while $x \in C$, and they undergo discrete jumps $x \mapsto G(x)$ when $x \in D$.

If one considers the parameter space where the architecture is fixed between growth stages and can be trained, then one can define a flow set C_q as

$$C_q = \{\theta \in \Theta_q \subseteq \mathbb{R}^{p_q}\}. \quad (1)$$

In this way, the flow F is given by gradient descent, for example SGD or Adam, over the weights with fixed architecture:

$$F_q = \dot{\theta} = -\nabla_{\theta} \mathcal{L}(f_{\theta,A}). \quad (2)$$

If the growth triggering function is described abstractly as $\chi_q : \Theta_q \rightarrow \mathbb{R}$, then it is possible to define

$$D_q = \{\theta \in \Theta_q : \chi_q(\theta) \geq 0\}. \quad (3)$$

At the same time, the jump map can be defined as

$$G_{q \rightarrow q^+} : \Theta_q \supseteq D_q \rightarrow \mathbb{R}^{p_q + \Delta p}. \quad (4)$$

So the training and growing system can be modeled following the Goebel–Sanfelice–Teel formalism [1].

$$\mathcal{H}_q : \begin{cases} \theta \in C_q = \Theta_q \subseteq \mathbb{R}^{p_q}, & \dot{\theta} = F_q(\theta) = -\nabla_{\theta} \mathcal{L}(f_{\theta,q}), \\ \theta \in D_q = \{\theta \in \Theta_q : \chi_q(\theta) \geq 0\}, & \theta^+ = G_{q \rightarrow q^+}(\theta) \in \Theta_{q^+} \subseteq \mathbb{R}^{p_q + \Delta p}. \end{cases} \quad (5)$$

1.2 Training stability

The continuous mode is nonlinear due to the neural network training dynamics. To study local behavior near a fixed architecture, we linearize the flow around an equilibrium point θ_q^* satisfying $F_q(\theta_q^*) = 0$. The complete first-order derivation is deferred to Appendix A.

For the gradient flow

$$\dot{\theta} = F_q(\theta) = -\nabla_{\theta} \mathcal{L}(f_{\theta,q}), \quad (6)$$

the first-order perturbation dynamics around θ_q^* is

$$\delta\dot{\theta} = -\nabla_{\theta}^2 \mathcal{L}(f_{\theta_q^*,q}) \delta\theta, \quad (7)$$

with $\delta\theta = \theta - \theta_q^*$. Hence, the corresponding linearized system matrix is

$$A_q = -\nabla_{\theta}^2 \mathcal{L}(f_{\theta_q^*,q}). \quad (8)$$

Therefore, local Hurwitz stability of the continuous mode depends on the spectrum of the Hessian at the equilibrium. However, due to the structural conditions of neural networks, it cannot be generally asserted that the linearized flow is Hurwitz, nor that the continuous modes are stable.

The objective is then to define a switching law that ensures that the entire switched system is exponentially stable for any switching signal, provided that the average dwell time is sufficiently large, as in [2].

$$\mathcal{S}_d[\tau_d] := \left\{ \left\{ (t_k, q_k) \right\}_{k \in \mathbb{N}} : \begin{array}{l} 0 \leq t_0 < t_1 < \dots, \\ t_{k+1} - t_k \geq \tau_d, \\ \theta(t_k) \in D_{q_k}, \\ \theta(t_k^+) = G_{q_k \rightarrow q_{k+1}}(\theta(t_k)) \end{array} \right\}, \quad \tau_d > 0. \quad (9)$$

For the linearized matrices A_q associated with each fixed architecture $q \in \{1, \dots, N\}$, assume, without loss of generality, that the first $r < N$ modes are unstable and that the remaining modes are Hurwitz. That is,

$$A_q \text{ is unstable for } q \in \{1, \dots, r\}, \quad A_q \text{ is Hurwitz for } q \in \{r+1, \dots, N\}.$$

Then, there exist scalars $\lambda_q > 0$ and $a_q \in \mathbb{R}$ such that, for all $t \geq 0$,

$$\|e^{A_q t}\| \leq \begin{cases} e^{a_q + \lambda_q t}, & q \in \{1, \dots, r\}, \\ e^{a_q - \lambda_q t}, & q \in \{r+1, \dots, N\}. \end{cases} \quad (10)$$

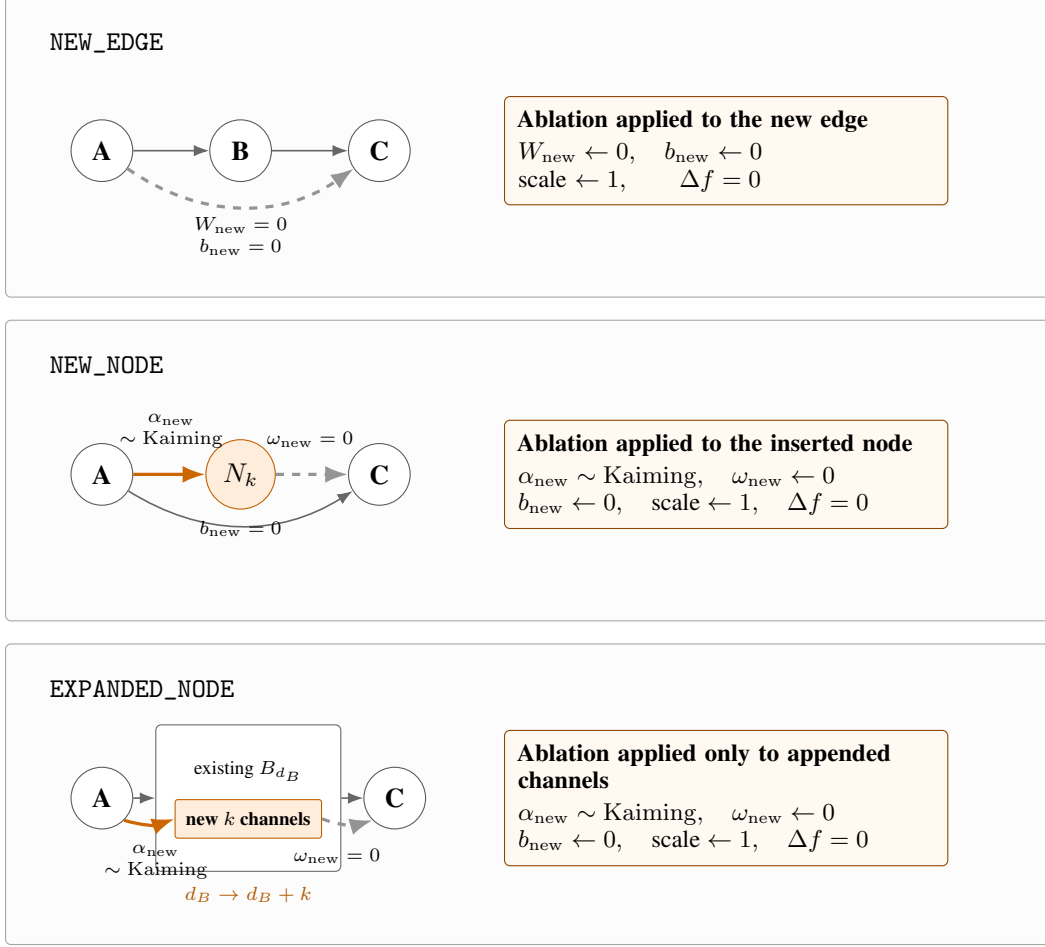


Figure 1: Initialization/scaling ablation shown independently for each selected action. Dashed gray arrows have zero weight at insertion; orange arrows use Kaiming initialization. In all cases the growth scale is fixed to one and the immediate function variation is $\Delta f = 0$.

2 Demeter initialization/scaling ablation

Once the shared pipeline has fixed the selected action, its location, and the number k of retained neurons, the ablation changes only the initialization and scaling applied to the new parameters.

These ablations are designed to isolate which part of the method affects growth and stability, while keeping the base pipeline unchanged whenever the corresponding option is not enabled in the configuration. The motivation is to test whether the local initialization can recover a training dynamic closer to the one observed with random initialization, and to better understand which mechanisms explain the observed results.

2.1 Ablated parameters

- **NEW_EDGE**: the parameters of the inserted edge are replaced by $W_{\text{new}} = 0$ and $b_{\text{new}} = 0$.
- **NEW_NODE**: the retained k neurons receive $\alpha_{\text{new}} \sim \text{Kaiming}$, while $\omega_{\text{new}} = 0$ and $b_{\text{new}} = 0$.
- **EXPANDED_NODE**: the existing channels are left unchanged. Only the appended k channels receive $\alpha_{\text{new}} \sim \text{Kaiming}$, $\omega_{\text{new}} = 0$, and $b_{\text{new}} = 0$.

For all three actions, the learned growth scale is replaced by a fixed value of one. The ablation therefore isolates the contribution of the initialization and scaling values without changing the previously selected structure.

2.2 Additional stabilization ablations

- **Init scaling ablation.** This variant evaluates an alternative initialization and scaling strategy for the new connections introduced during growth. The goal is to compare the original local initialization against a setting where the scale of the newly added DAG branches is controlled explicitly.
- **Linear warmup scheduler.** This variant introduces a linear learning-rate warmup for a fixed number of epochs before switching to the normal scheduler dynamics. It tests whether a smoother optimization start improves stability after structural changes. Early stopping is disabled temporarily during the warmup period.
- **Global scheduler 350.** This variant keeps a fixed global scheduler horizon, for example 350 epochs, without restarting the scheduler between training phases. Conceptually, it tests whether a continuous notion of training time improves the comparison between progressively grown models.
- **Post-growth scheduler.** This variant disables learning-rate scheduling during the growth phases. Scheduling starts only after structural growth has finished, which separates architecture optimization from the final optimization of the fixed model.
- **Growth label smoothing.** This variant applies label smoothing only while the model is growing. Once growth ends, label smoothing is disabled and the final training phase uses the standard loss. The intent is to regularize growth decisions without biasing the final fitting stage.
- **Noisy pre-activation gradients.** This variant adds Gaussian noise to the functional gradients used by Demeter to estimate new neurons. The objective is to prevent the new neuron from approximating the gradient too exactly, which may otherwise encourage representational collapse. The noise affects growth estimation only, not standard training.
- **Variance transfer / vt_constraint_old_shape.** This variant rescales new connections using statistics from existing weights to preserve the expected variance of activations. It is intended to avoid abrupt changes in forward-pass scale, especially when several nodes merge into the same downstream node.

2.3 Results

The plots and detailed run comparisons for this ablation are available in the following W&B report: [demeter-init-scaling-ablation](#).

3 Training and growing with FGD

Considering the problem of convergence toward local minima during the training stages based on parametric gradient-descent strategies, we use [3] as a reference for exploring how training could be implemented through functional gradient descent within a dynamical-system formulation. The objective is to adjust the growth process so that it remains compatible with the constraints of the functional-gradient framework: a relative error bound, convergence to a stationary point, and convergence to a global minimizer.

Let $f_\theta : \mathcal{X} \rightarrow \mathbb{R}^c$ be a neural network for a batch $\mathcal{B}$ as in (11). We propose to use the empirical functional space in (13), where the empirical measure is given by (12). The norm is defined in (14).

$$f_\theta : \mathcal{X} \rightarrow \mathbb{R}^c, \quad \mathcal{B} = \{(x_i, y_i)\}_{i=1}^m. \quad (11)$$

$$P_{\mathcal{B}} = \frac{1}{m} \sum_{i=1}^m \delta_{x_i}. \quad (12)$$

$$\mathcal{V}_{\mathcal{B}} = L^2(P_{\mathcal{B}}; \mathbb{R}^c) \simeq \mathbb{R}^{m \times c}. \quad (13)$$

$$\|v\|_{L^2(P_{\mathcal{B}})}^2 = \frac{1}{m} \sum_{i=1}^m \|v(x_i)\|_2^2, \quad v \in \mathcal{V}_{\mathcal{B}}. \quad (14)$$

In this scenario, the loss function is defined in (15). This is convex with respect to the output $z_{\theta, \mathcal{B}} = f_\theta(\mathcal{B})$ defined in (16).

$$\mathcal{L}(f_\theta) = \frac{1}{2} \|f_\theta - y\|_{L^2(P_{\mathcal{B}})}^2 = \frac{1}{2m} \sum_{i=1}^m \|f_\theta(x_i) - y_i\|_2^2. \quad (15)$$

$$z_{\theta, \mathcal{B}} = f_\theta(\mathcal{B}) = (f_\theta(x_1), \dots, f_\theta(x_m)) \in \mathbb{R}^{m \times c}. \quad (16)$$

Following the DEMETER definition, the structure of a network can only represent the function class in (17), and the functional directions that it can reproduce are contained in the tangent space in (18).

$$\mathcal{F}_A = \{f_\theta : \theta \in \Theta_A\}. \quad (17)$$

$$T_A = \{J_\theta \delta\theta : \delta\theta \in \mathbb{R}^p\}. \quad (18)$$

Then, although the ideal direction is (19), the best direction that can be reached with the structure of the network is (20).

$$v_\nabla = -\nabla \mathcal{L}(f_\theta) = y - f_\theta. \quad (19)$$

$$v^* = \text{Proj}_{T_A}(v_\nabla). \quad (20)$$

where $\delta\theta^*$ is defined by (21), and v^* is given by (22). The expressivity bottleneck is described as (23).

$$\delta\theta^* = \arg \min_{\delta\theta \in \mathbb{R}^p} \|J_\theta \delta\theta - v_\nabla\|_{L^2(P_{\mathcal{B}})}^2. \quad (21)$$

$$v^* = J_\theta \delta\theta^*. \quad (22)$$

$$v_{\perp} = v_{\nabla} - v^{\star}. \quad (23)$$

The general optimization problem can be summarized starting from an arbitrary structure as in (24). In accordance with [3], one can estimate an approximation to the functional gradient $g_{t,\theta}$, as written in (25).

$$f_{t+1,\theta}^{\text{ideal}} = f_{t,\theta}^{\text{ideal}} - \eta \nabla \mathcal{L}(f_{t,\theta}^{\text{ideal}}). \quad (24)$$

$$f_{t+1,\theta} = f_{t,\theta} - \eta g_{t,\theta}. \quad (25)$$

It is known that the directions of the network are limited by its tangent space; therefore, the best way to approximate the functional gradient in a structure is defined by the direction inside the tangent space of the optimal modification of the network parameters, as written in (26).

$$v^{\star} = J_{\theta} \delta \theta^{\star} = g_{t,\theta}. \quad (26)$$

Under this definition of the approximation function, we propose to state the assumptions introduced by the author, which finally lead to convergence to a single point that is a global minimizer and not only a local one. The first of these assumptions is the smoothness condition in (27). In the particular case of the proposed loss function, the smoothness constant is equal to 1; the proof is given in Appendix B.

$$\mathcal{L}(f) \leq \mathcal{L}(f') + D\mathcal{L}(f'; f - f') + \frac{1}{2} \|f - f'\|_{L^2(P_{\mathcal{B}})}^2. \quad (27)$$

The next assumption is that the loss descends in $\mathcal{V}_{\mathcal{B}}$. For the proposed loss, the corresponding constant is equal to 1, so for every $f \in \mathcal{V}_{\mathcal{B}}$,

$$D\mathcal{L}(f; \nabla \mathcal{L}(f)) \geq \|\nabla \mathcal{L}(f)\|_{L^2(P_{\mathcal{B}})}^2. \quad (28)$$

The gradient compatibility condition also has constant equal to 1 in this empirical Hilbert space, and therefore for every $f \in \mathcal{V}_{\mathcal{B}}$,

$$\|D\mathcal{L}(f; \cdot)\|_{\mathcal{V}_{\mathcal{B}}^*} \leq \|\nabla \mathcal{L}(f)\|_{L^2(P_{\mathcal{B}})}. \quad (29)$$

Both identities follow because the Banach space used in the assumptions is the same empirical Hilbert functional space $\mathcal{V}_{\mathcal{B}}$.

Starting from the approximation defined in (26), one obtains the relative error with respect to the direction implemented by the current structure in (30).

$$\text{RelErr}(g_{t,\theta}, \nabla \mathcal{L}(f_{t,\theta})) = \frac{\|g_{t,\theta} - \nabla \mathcal{L}(f_{t,\theta})\|_{L^2(P_{\mathcal{B}})}}{\|g_{t,\theta}\|_{L^2(P_{\mathcal{B}})}} = \frac{\|v_{\perp}\|_{L^2(P_{\mathcal{B}})}}{\|v^{\star}\|_{L^2(P_{\mathcal{B}})}}. \quad (30)$$

Let

$$\varepsilon_t = \text{RelErr}(g_{t,\theta}, \nabla \mathcal{L}(f_{t,\theta})). \quad (31)$$

Using the sufficient descent lemma with the values already fixed for the proposed loss, namely smoothness constant equal to 1 and descent and compatibility constants equal to 1, the proposed train-and-grow system satisfies (32).

$$\mathcal{L}(f_{t+1,\theta}) \leq \mathcal{L}(f_{t,\theta}) - \eta \left[1 - \frac{\eta}{2} - \left(1 + \frac{3}{2}\eta \right) \frac{\varepsilon_t}{1 - \varepsilon_t} \right] \|\nabla \mathcal{L}(f_{t,\theta})\|_{L^2(P_{\mathcal{B}})}^2. \quad (32)$$

For gradient descent to guarantee convergence to a stationary point, it is necessary that the relative error remains below the admissible threshold in (33). After replacing the constants already fixed above, this threshold is $\varepsilon < 1/2$.

$$\text{RelErr}(g_{t,\theta}, \nabla \mathcal{L}(f_{t,\theta})) \leq \varepsilon, \quad t = 0, \dots, T-1, \quad 0 \leq \varepsilon < \frac{1}{2}. \quad (33)$$

Then, for any learning rate satisfying (34), the stationary-point bound in (35) follows.

$$\eta \leq \frac{2(1 - 2\varepsilon)}{2\varepsilon + 1}. \quad (34)$$

$$\min_{t=0, \dots, T-1} \|\nabla \mathcal{L}(f_{t,\theta})\|_{L^2(P_B)}^2 \leq \frac{\mathcal{L}(f_{0,\theta}) - \mathcal{L}^*}{T\eta r}. \quad (35)$$

Moreover, under the conditions of the previous propositions, if the learning rate satisfies the strict condition in (36), one can ensure that after T iterations the method converges to a global minimizer whenever the contraction in (37) is guaranteed.

$$\eta < \frac{2(1 - 2\varepsilon)}{2\varepsilon + 1}. \quad (36)$$

$$\mathcal{L}(f_{T,\theta}) - \mathcal{L}^* \leq (1 - 2\eta\mu r)^T (\mathcal{L}(f_{0,\theta}) - \mathcal{L}^*). \quad (37)$$

For the finite-dimensional batch-space formulation, the PL constant used in this contraction is $\mu_B = 1$; see Appendix D.

Thus, the complete train-and-grow system can be interpreted as an adaptive gradient-descent process over a sequence of network structures. Starting from an arbitrary initial structure A_s , the parameters are trained while the projected direction $g_{t,\theta} \in T_{A_s}$ remains a sufficiently accurate approximation of the functional gradient. The structure is kept fixed when the relative error in (33), the sufficient descent condition in (32), and the convergence bounds in (35) and (37) are compatible with the current learning rate.

Growth is triggered when the current structure does not provide enough expressivity to certify the functional-gradient descent requirements. In practice, this occurs whenever at least one of the three FGD conditions is not satisfied: the relative-error bound in (33), the stationary-point bound in (35), or the global-convergence bound in (37). This rule is summarized in (38).

$$\text{grow if} \quad \neg (\mathcal{C}_{\text{rel}} \wedge \mathcal{C}_{\text{stat}} \wedge \mathcal{C}_{\text{glob}}), \quad (38)$$

where $\mathcal{C}_{\text{rel}}$ denotes the relative-error condition, $\mathcal{C}_{\text{stat}}$ denotes convergence to a stationary point, and $\mathcal{C}_{\text{glob}}$ denotes convergence to a global minimizer.

When growth is required, the DEMETER restricted strategy is used to generate local alternatives around the current bottleneck: increasing the width of a layer, adding a node, or adding an admissible edge between existing nodes. Each candidate is locally optimized and then filtered by the same FGD constraints. Among the admissible candidates, the next structure is selected as the one that most reduces the expressivity bottleneck, as described in (39).

$$A_{s+1} \in \arg \min_{A' \in \mathcal{C}_{\text{adm}}(A_s)} \|v_{\perp}(A')\|_{L^2(P_B)}. \quad (39)$$

After the new structure A_{s+1} is selected, the parameters inherited from A_s are kept, the new parameters are locally initialized, and the same training-and-certification cycle is repeated. Figure 2 summarizes this epoch-wise transition from one structure to the next.

3.1 FGD formulation with RKHS

We propose to use an RKHS in order to give a suitable functional setting for the MSE. First, we define the space where the functional gradient of the loss lives as

$$\mathcal{H} = \mathcal{H}_K^c. \quad (40)$$

Equivalently, this means that the vector-valued RKHS is the product of c copies of the scalar RKHS:

$$\mathcal{H}_K^c = \underbrace{\mathcal{H}_K \times \dots \times \mathcal{H}_K}_{c \text{ times}}. \quad (41)$$

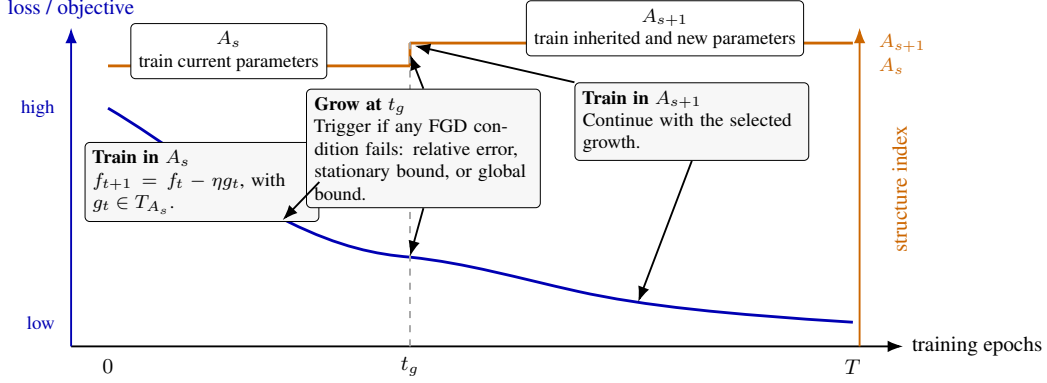


Figure 2: Epoch-wise train-and-grow trajectory for the FGD formulation. The blue curve uses the left axis and represents the continuous loss evolution, while the orange step uses the right axis and represents the discrete network structure. At the growth epoch t_g , the current structure is changed only if the FGD criteria fail, after which training continues on the selected admissible structure.

Under this choice, the RKHS gradient of the batch loss belongs to (40):

$$\nabla_{\mathcal{H}} \mathcal{L}_{\mathcal{B}}(f) \in \mathcal{H}. \quad (42)$$

It is then necessary to define a larger Banach space in which both the network and its tangent directions can be represented. We use

$$\mathcal{B}_{\infty} = L^{\infty}(\mathcal{X}; \mathbb{R}^c). \quad (43)$$

The norm in this Banach space is

$$\|u\|_{\mathcal{B}_{\infty}} = \text{ess sup}_{x \in \mathcal{X}} \|u(x)\|_2, \quad u \in \mathcal{B}_{\infty}. \quad (44)$$

On a finite batch, we denote the induced restriction of this norm by $\|\cdot\|_{\mathcal{B}}$:

$$\|U_{\mathcal{B}}\|_{\mathcal{B}} = \max_{1 \leq i \leq m} \|U_i\|_2, \quad U_{\mathcal{B}} \in \mathbb{R}^{m \times c}. \quad (45)$$

Then the RKHS gradient and the tangent directions are both embedded in $\mathcal{B}_{\infty}$ and are compared with the norm in (44):

$$\nabla_{\mathcal{H}} \mathcal{L}_{\mathcal{B}}(f) \in \mathcal{H} \subset \mathcal{B}_{\infty}, \quad J_{\theta} \delta \theta \in \mathcal{B}_{\infty}. \quad (46)$$

This allows the functional gradient to be compared with the directions of the tangent space, because both objects belong to the same Banach space.

The RKHS functional gradient of the batch loss is given by (47); its derivation is included in Appendix E.

$$\nabla_{\mathcal{H}} \mathcal{L}_{\mathcal{B}}(f)(x) = \frac{1}{m} \sum_{i=1}^m K(x_i, x) (f(x_i) - y_i). \quad (47)$$

For compact batch notation, define the uppercase vectors

$$F_{\theta, \mathcal{B}} = (f_{\theta}(x_1), \dots, f_{\theta}(x_m)), \quad Y_{\mathcal{B}} = (y_1, \dots, y_m). \quad (48)$$

The kernel batch operator is defined by its action on any $R_{\mathcal{B}} = (R_1, \dots, R_m) \in \mathbb{R}^{m \times c}$:

$$(K_{\mathcal{B}} R_{\mathcal{B}})_i = \sum_{j=1}^m K(x_j, x_i) R_j, \quad i = 1, \dots, m. \quad (49)$$

With these definitions, the RKHS descent direction evaluated on the batch is

$$-\nabla_{\mathcal{H}} \mathcal{L}_{\mathcal{B}}(f_{\theta}) = \frac{1}{m} K_{\mathcal{B}} (Y_{\mathcal{B}} - F_{\theta, \mathcal{B}}). \quad (50)$$

Therefore, the best direction reachable by the current network structure is

$$\delta\theta_{\mathcal{H},\mathcal{B}}^* = \arg \min_{\delta\theta \in \mathbb{R}^p} \left\| J_{\theta,\mathcal{B}} \delta\theta - \frac{1}{m} K_{\mathcal{B}} (Y_{\mathcal{B}} - F_{\theta,\mathcal{B}}) \right\|_{\mathcal{B}}. \quad (51)$$

The corresponding reproducible descent direction is

$$v_{\mathcal{H},\mathcal{B}}^* = J_{\theta,\mathcal{B}} \delta\theta_{\mathcal{H},\mathcal{B}}^*. \quad (52)$$

The approximation residual in the Banach space is then

$$r_{\mathcal{H},\mathcal{B}} = \frac{1}{m} K_{\mathcal{B}} (Y_{\mathcal{B}} - F_{\theta,\mathcal{B}}) - v_{\mathcal{H},\mathcal{B}}^*. \quad (53)$$

Thus, the relative error can be written directly in terms of the RKHS functional gradient:

$$\text{RelErr}_{\mathcal{H},\mathcal{B}} = \frac{\|r_{\mathcal{H},\mathcal{B}}\|_{\mathcal{B}}}{\|v_{\mathcal{H},\mathcal{B}}^*\|_{\mathcal{B}}} = \frac{\|J_{\theta,\mathcal{B}} \delta\theta_{\mathcal{H},\mathcal{B}}^* - \frac{1}{m} K_{\mathcal{B}} (Y_{\mathcal{B}} - F_{\theta,\mathcal{B}})\|_{\mathcal{B}}}{\|J_{\theta,\mathcal{B}} \delta\theta_{\mathcal{H},\mathcal{B}}^*\|_{\mathcal{B}}}. \quad (54)$$

The conditions required to obtain convergence to a unique global minimizer under this FGD formulation now depend strongly on the selected kernel. The kernel should therefore be fixed before training and should satisfy three requirements. First, it must define a valid RKHS, so it must be positive definite. Second, it must be bounded, so that the batch estimates remain controlled. Third, for every admissible batch, its Gram matrix must be positive definite, since the constants below depend on the smallest eigenvalue of that matrix. In compact form, the desired properties are

$$K_{\mathcal{B}} \succ 0, \quad \lambda_{\mathcal{B}} = \lambda_{\min}(K_{\mathcal{B}}) > 0, \quad \kappa = \sup_{x, x' \in \mathcal{X}} |K(x, x')| < \infty. \quad (55)$$

The kernel should also encode a meaningful similarity in feature space, because the ideal RKHS direction is obtained by applying the Gram matrix to the residual vector. Thus, the kernel determines how residual information is shared across nearby batch points before being approximated by the tangent directions of the network.

As a baseline kernel satisfying these requirements, we use the normalized RBF kernel

$$K_{\sigma}(x, x') = \exp \left(-\frac{\|\varphi(x) - \varphi(x')\|_2^2}{2\sigma^2} \right), \quad \sigma > 0. \quad (56)$$

The norm inside the kernel represents a distance in the feature space. Hence, if two points are close in this space, then

$$K_{\sigma}(x, x') \approx 1. \quad (57)$$

If they are far apart, then

$$K_{\sigma}(x, x') \approx 0. \quad (58)$$

Moreover, for the RBF kernel,

$$0 < K_{\sigma}(x, x') \leq 1. \quad (59)$$

Therefore,

$$\kappa = \sup_{x, x' \in \mathcal{X}} |K_{\sigma}(x, x')| = 1. \quad (60)$$

The kernel matrix over the batch is

$$K_{\mathcal{B}} = [K_{\sigma}(x_i, x_j)]_{i,j=1}^m, \quad K_{\mathcal{B}} \in \mathbb{R}^{m \times m}. \quad (61)$$

The admissible batches are assumed to contain pairwise distinct feature points, namely

$$\varphi(x_1), \dots, \varphi(x_m) \quad (62)$$

are pairwise distinct. Under this condition, the RBF kernel is strictly positive definite on the batch and therefore

$$K_{\mathcal{B}} \succ 0. \quad (63)$$

Hence its smallest eigenvalue is well-defined and positive:

$$\lambda_{\mathcal{B}} = \lambda_{\min}(K_{\mathcal{B}}) > 0. \quad (64)$$

Appendix F proves the same 1-smoothness statement with respect to the Banach batch norm $\|\cdot\|_{\mathcal{B}}$. To keep the notation consistent with the batch quantities introduced above, define the residual vector

$$R_{\mathcal{B}} = F_{\theta, \mathcal{B}} - Y_{\mathcal{B}} = (r_1, \dots, r_m), \quad r_i = f_{\theta}(x_i) - y_i. \quad (65)$$

Using the reproducing-property derivation in Appendix E, for any RKHS perturbation $h \in \mathcal{H}$ the directional derivative can be written as

$$D\mathcal{L}_{\mathcal{B}}(f_{\theta}; h) = \left\langle \frac{1}{m} \sum_{i=1}^m K(x_i, \cdot) r_i, h \right\rangle_{\mathcal{H}} = \frac{1}{m} \sum_{i=1}^m \langle r_i, h(x_i) \rangle_{\mathbb{R}^c}. \quad (66)$$

Assume now that $\|h\|_{\mathcal{B}} \leq 1$. Then, by the definition of the batch norm, $\|h(x_i)\|_2 \leq 1$ for every $i = 1, \dots, m$. Hence,

$$|D\mathcal{L}_{\mathcal{B}}(f_{\theta}; h)| \leq \frac{1}{m} \sum_{i=1}^m |\langle r_i, h(x_i) \rangle_{\mathbb{R}^c}| \leq \frac{1}{m} \sum_{i=1}^m \|r_i\|_2. \quad (67)$$

Taking the supremum over all perturbations satisfying $\|h\|_{\mathcal{B}} \leq 1$ gives the induced dual-norm bound

$$\|D\mathcal{L}_{\mathcal{B}}(f_{\theta}; \cdot)\|_{\mathcal{B},*} = \sup_{\|h\|_{\mathcal{B}} \leq 1} |D\mathcal{L}_{\mathcal{B}}(f_{\theta}; h)| \leq \frac{1}{m} \sum_{i=1}^m \|r_i\|_2. \quad (68)$$

The kernel-dependent compatibility bound is proved in Appendix G. With the notation above, it gives

$$\|D\mathcal{L}_{\mathcal{B}}(f_{\theta}; \cdot)\|_{\mathcal{B},*} \leq \beta_{\mathcal{K}} \|\nabla_{\mathcal{H}} \mathcal{L}_{\mathcal{B}}(f_{\theta})\|_{\mathcal{B}}, \quad \beta_{\mathcal{K}} = \frac{m}{\lambda_{\mathcal{B}}}. \quad (69)$$

The corresponding RKHS descent constant is derived in Appendix H. It gives

$$D\mathcal{L}_{\mathcal{B}}(f_{\theta}; \nabla_{\mathcal{H}} \mathcal{L}_{\mathcal{B}}(f_{\theta})) \geq \alpha_{\mathcal{K}} \|\nabla_{\mathcal{H}} \mathcal{L}_{\mathcal{B}}(f_{\theta})\|_{\mathcal{B}}^2, \quad \alpha_{\mathcal{K}} = \frac{\lambda_{\mathcal{B}}}{m}. \quad (70)$$

Substituting the Banach smoothness constant $K_{\text{sm}, \mathcal{B}} = 1$, $\alpha_{\mathcal{K}} = \lambda_{\mathcal{B}}/m$, and $\beta_{\mathcal{K}} = m/\lambda_{\mathcal{B}}$ into the sufficient descent lemma gives the RKHS-specific descent inequality. To keep the expression readable, denote

$$\varepsilon_{\mathcal{H}, \mathcal{B}, t} = \text{RelErr}_{\mathcal{H}, \mathcal{B}}(g_{t, \theta}, \nabla_{\mathcal{H}} \mathcal{L}_{\mathcal{B}}(f_{t, \theta})). \quad (71)$$

Then the descent coefficient is

$$\Gamma_{\mathcal{H}, \mathcal{B}, t} = \frac{\lambda_{\mathcal{B}}}{m} - \frac{\eta}{2} - \left(\frac{m}{\lambda_{\mathcal{B}}} + \frac{3}{2}\eta \right) \frac{\varepsilon_{\mathcal{H}, \mathcal{B}, t}}{1 - \varepsilon_{\mathcal{H}, \mathcal{B}, t}}, \quad (72)$$

and therefore

$$\mathcal{L}_{\mathcal{B}}(f_{t+1, \theta}) \leq \mathcal{L}_{\mathcal{B}}(f_{t, \theta}) - \eta \Gamma_{\mathcal{H}, \mathcal{B}, t} \|\nabla_{\mathcal{H}} \mathcal{L}_{\mathcal{B}}(f_{t, \theta})\|_{\mathcal{B}}^2. \quad (73)$$

For functional gradient descent in the RKHS setting to guarantee convergence to a stationary point, the Banach relative error must remain below the admissible threshold

$$\varepsilon_{\mathcal{H}, \mathcal{B}, t} \leq \varepsilon, \quad t = 0, \dots, T-1, \quad 0 \leq \varepsilon < \min \left\{ \frac{1}{2}, \frac{\lambda_{\mathcal{B}}^2}{\lambda_{\mathcal{B}}^2 + m^2} \right\}. \quad (74)$$

In that case, any learning rate satisfying

$$\eta \leq \frac{2 \left[\frac{\lambda_{\mathcal{B}}}{m} - \left(\frac{\lambda_{\mathcal{B}}}{m} + \frac{m}{\lambda_{\mathcal{B}}} \right) \varepsilon \right]}{1 + 2\varepsilon} \quad (75)$$

gives a positive Banach descent coefficient

$$r_{\mathcal{H}, \mathcal{B}} = \frac{\lambda_{\mathcal{B}}}{m} - \frac{\eta}{2} - \left(\frac{m}{\lambda_{\mathcal{B}}} + \frac{3}{2}\eta \right) \frac{\varepsilon}{1 - \varepsilon} > 0. \quad (76)$$

Therefore, the stationary-point bound is written in the Banach batch norm as

$$\min_{t=0, \dots, T-1} \|\nabla_{\mathcal{H}} \mathcal{L}_{\mathcal{B}}(f_{t, \theta})\|_{\mathcal{B}}^2 \leq \frac{\mathcal{L}_{\mathcal{B}}(f_{0, \theta}) - \mathcal{L}_{\mathcal{B}}^*}{T\eta r_{\mathcal{H}, \mathcal{B}}}. \quad (77)$$

If the batch is interpolable, the corresponding RKHS PL condition in the Banach batch norm is verified in Appendix I, which gives $\mu_{\mathcal{K}} = 1/m$. Under this condition, the RKHS formulation satisfies the global convergence estimate

$$\mathcal{L}_{\mathcal{B}}(f_{T, \theta}) - \mathcal{L}_{\mathcal{B}}^* \leq (1 - 2\eta\mu_{\mathcal{K}}r_{\mathcal{H}, \mathcal{B}})^T (\mathcal{L}_{\mathcal{B}}(f_{0, \theta}) - \mathcal{L}_{\mathcal{B}}^*). \quad (78)$$

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A First-order derivation for the training stability analysis

This appendix provides the first-order derivation used in Section 1.2 to obtain the local linear model for the continuous training dynamics.

We linearize the dynamics around an equilibrium point θ_q^* . The point θ_q^* is an equilibrium if

$$F_q(\theta_q^*) = 0, \quad (79)$$

where the gradient vanishes. If a small perturbation $\delta\theta = \theta - \theta_q^*$ is introduced, then

$$\dot{\delta\theta} = \dot{\theta}. \quad (80)$$

Since

$$\dot{\theta} = F_q(\theta), \quad (81)$$

one obtains

$$\dot{\delta\theta} = F_q(\theta_q^* + \delta\theta). \quad (82)$$

Applying the first-order expansion around θ_q^* gives

$$F_q(\theta_q^* + \delta\theta) = F_q(\theta_q^*) + DF_q(\theta_q^*)\delta\theta + o(\|\delta\theta\|). \quad (83)$$

Since θ_q^* is an equilibrium,

$$F_q(\theta_q^*) = 0. \quad (84)$$

Therefore, the expansion becomes

$$F_q(\theta_q^* + \delta\theta) = DF_q(\theta_q^*)\delta\theta + o(\|\delta\theta\|). \quad (85)$$

Then, the perturbed dynamics is given by

$$\dot{\delta\theta} = DF_q(\theta_q^*)\delta\theta + o(\|\delta\theta\|). \quad (86)$$

For the gradient flow associated with the loss function, one has

$$DF_q(\theta) = -\nabla_{\theta}^2 \mathcal{L}(f_{\theta,q}). \quad (87)$$

Thus, the linearized dynamics around θ_q^* is

$$\dot{\delta\theta} = -\nabla_{\theta}^2 \mathcal{L}(f_{\theta_q^*,q})\delta\theta. \quad (88)$$

Hence,

$$A_q = -\nabla_{\theta}^2 \mathcal{L}(f_{\theta_q^*,q}). \quad (89)$$

B 1-smoothness for the FGD batch loss

This appendix verifies the 1-smoothness condition in (27) for the quadratic batch loss used in Section 3.

For a batch $\mathcal{B}$, define

$$\mathcal{L}_{\mathcal{B}}(f) = \frac{1}{2} \|f - y\|_{L^2(P_{\mathcal{B}})}^2. \quad (90)$$

Equivalently,

$$\mathcal{L}_{\mathcal{B}}(f) = \frac{1}{2m} \sum_{i=1}^m \|f(x_i) - y_i\|_2^2. \quad (91)$$

We want to verify the condition

$$\mathcal{L}_{\mathcal{B}}(f) \leq \mathcal{L}_{\mathcal{B}}(f') + D\mathcal{L}_{\mathcal{B}}(f'; f - f') + \frac{1}{2} \|f - f'\|_{L^2(P_{\mathcal{B}})}^2. \quad (92)$$

For the network function f_θ , let

$$h = f - f'. \quad (93)$$

Then the directional derivative is

$$D\mathcal{L}_B(f'; h) = \langle f' - y, h \rangle_{L^2(P_B)}. \quad (94)$$

That is,

$$D\mathcal{L}_B(f'; f - f') = \frac{1}{m} \sum_{i=1}^m \langle f'(x_i) - y_i, f(x_i) - f'(x_i) \rangle. \quad (95)$$

Now expand the loss:

$$\mathcal{L}_B(f) = \frac{1}{2} \|f - y\|_{L^2(P_B)}^2. \quad (96)$$

Since

$$f - y = (f' - y) + (f - f'), \quad (97)$$

we obtain

$$\|f - y\|_{L^2(P_B)}^2 = \|(f' - y) + (f - f')\|_{L^2(P_B)}^2. \quad (98)$$

Using

$$\|a + b\|^2 = \|a\|^2 + 2\langle a, b \rangle + \|b\|^2, \quad (99)$$

we get

$$\|f - y\|_{L^2(P_B)}^2 = \|f' - y\|_{L^2(P_B)}^2 + 2\langle f' - y, f - f' \rangle_{L^2(P_B)} + \|f - f'\|_{L^2(P_B)}^2. \quad (100)$$

Multiplying by 1/2 gives

$$\mathcal{L}_B(f) = \mathcal{L}_B(f') + \langle f' - y, f - f' \rangle_{L^2(P_B)} + \frac{1}{2} \|f - f'\|_{L^2(P_B)}^2. \quad (101)$$

Using (94), this becomes

$$\mathcal{L}_B(f) = \mathcal{L}_B(f') + D\mathcal{L}_B(f'; f - f') + \frac{1}{2} \|f - f'\|_{L^2(P_B)}^2. \quad (102)$$

Thus, the exact expansion in (102) implies the required 1-smoothness condition in (92):

$$\mathcal{L}_B(f) \leq \mathcal{L}_B(f') + D\mathcal{L}_B(f'; f - f') + \frac{1}{2} \|f - f'\|_{L^2(P_B)}^2. \quad (103)$$

C Batch as a finite-dimensional function space

Before introducing the RKHS construction, one can use the batch itself as the finite-dimensional function space. In that case, each function is represented only through its values on the batch,

$$F_B = (f(x_1), \dots, f(x_m)) \in \mathbb{R}^{m \times c}, \quad Y_B = (y_1, \dots, y_m). \quad (104)$$

The empirical Hilbert structure is

$$\langle U_B, V_B \rangle_{L^2(P_B)} = \frac{1}{m} \sum_{i=1}^m \langle u_i, v_i \rangle_{\mathbb{R}^c}, \quad \|U_B\|_{L^2(P_B)}^2 = \frac{1}{m} \sum_{i=1}^m \|u_i\|_2^2. \quad (105)$$

For the Banach formulation used in the descent assumptions, the same batch space is equipped with the maximum pointwise norm

$$\|U_B\|_B = \max_{1 \leq i \leq m} \|u_i\|_2. \quad (106)$$

Let

$$C_B = F_B - Y_B = (c_1, \dots, c_m), \quad c_i = f(x_i) - y_i. \quad (107)$$

Then the quadratic batch loss is

$$\mathcal{L}_B(f) = \frac{1}{2m} \|C_B\|_F^2. \quad (108)$$

For any perturbation $H_B = (h_1, \dots, h_m)$, the directional derivative is

$$D\mathcal{L}_B(f; H_B) = \frac{1}{m} \sum_{i=1}^m \langle c_i, h_i \rangle_{\mathbb{R}^c}. \quad (109)$$

Thus, when the Hilbert structure (105) is used, the functional gradient is simply the residual vector C_B . When the Banach norm (106) is used, the same directional derivative is controlled through its induced dual norm.

D PL condition in the finite-dimensional batch space

In the finite-dimensional batch space, no kernel is involved. Therefore, we write the corresponding PL constant as $\mu_{\mathcal{B}}$. We verify

$$\mathcal{L}_{\mathcal{B}}(f) - \mathcal{L}_{\mathcal{B}}^* \leq \frac{1}{2\mu_{\mathcal{B}}} \|D\mathcal{L}_{\mathcal{B}}(f; \cdot)\|_{\mathcal{V}_{\mathcal{B}}^*}^2. \quad (110)$$

If the batch can be interpolated, then $\mathcal{L}_{\mathcal{B}}^* = 0$. From (108),

$$\mathcal{L}_{\mathcal{B}}(f) = \frac{1}{2m} \|C_{\mathcal{B}}\|_F^2. \quad (111)$$

Using the empirical Hilbert structure in (105), the directional derivative can be written as

$$D\mathcal{L}_{\mathcal{B}}(f; H_{\mathcal{B}}) = \langle C_{\mathcal{B}}, H_{\mathcal{B}} \rangle_{L^2(P_{\mathcal{B}})}. \quad (112)$$

Hence

$$\|D\mathcal{L}_{\mathcal{B}}(f; \cdot)\|_{\mathcal{V}_{\mathcal{B}}^*} = \|C_{\mathcal{B}}\|_{L^2(P_{\mathcal{B}})}. \quad (113)$$

Therefore,

$$\|D\mathcal{L}_{\mathcal{B}}(f; \cdot)\|_{\mathcal{V}_{\mathcal{B}}^*}^2 = \frac{1}{m} \|C_{\mathcal{B}}\|_F^2. \quad (114)$$

Substituting this identity into (111) gives

$$\mathcal{L}_{\mathcal{B}}(f) = \frac{1}{2} \|D\mathcal{L}_{\mathcal{B}}(f; \cdot)\|_{\mathcal{V}_{\mathcal{B}}^*}^2. \quad (115)$$

Comparing (115) with (110) yields

$$\frac{1}{2\mu_{\mathcal{B}}} = \frac{1}{2}, \quad \mu_{\mathcal{B}} = 1. \quad (116)$$

E Functional gradient in RKHS

Let $v \in \mathcal{H}$ be an infinitesimal functional perturbation. For the batch loss, the perturbed objective satisfies

$$\mathcal{L}_{\mathcal{B}}(f + \epsilon v) - \mathcal{L}_{\mathcal{B}}(f) = \frac{1}{2m} \sum_{i=1}^m \left[\|f(x_i) + \epsilon v(x_i) - y_i\|_2^2 - \|f(x_i) - y_i\|_2^2 \right]. \quad (117)$$

Using the pointwise loss

$$\ell(u, y_i) = \frac{1}{2} \|u - y_i\|_2^2, \quad (118)$$

the first-order Taylor expansion gives

$$\ell(f(x_i) + \epsilon v(x_i), y_i) - \ell(f(x_i), y_i) = \epsilon \left\langle \nabla_u \ell(u, y_i) \Big|_{u=f(x_i)}, v(x_i) \right\rangle + O(\epsilon^2 \|v(x_i)\|_2^2). \quad (119)$$

For the MSE loss,

$$\nabla_u \ell(u, y_i) \Big|_{u=f(x_i)} = f(x_i) - y_i. \quad (120)$$

Substituting (120) into (119) yields

$$\mathcal{L}_{\mathcal{B}}(f + \epsilon v) - \mathcal{L}_{\mathcal{B}}(f) = \frac{\epsilon}{m} \sum_{i=1}^m \langle f(x_i) - y_i, v(x_i) \rangle + O(\epsilon^2). \quad (121)$$

Therefore, after dividing by ϵ and taking the limit $\epsilon \rightarrow 0$,

$$D\mathcal{L}_{\mathcal{B}}(f; v) = \frac{1}{m} \sum_{i=1}^m \langle f(x_i) - y_i, v(x_i) \rangle. \quad (122)$$

If $\mathcal{H} = \mathcal{H}_K^c$ is the vector-valued RKHS used in Section 3, the reproducing property gives

$$\langle K(x_i, \cdot) (f(x_i) - y_i), v \rangle_{\mathcal{H}} = \langle f(x_i) - y_i, v(x_i) \rangle_{\mathbb{R}^c}. \quad (123)$$

Indeed, if $r_i = f(x_i) - y_i = (r_{i,1}, \dots, r_{i,c})$ and $v = (v_1, \dots, v_c)$, then

$$K(x_i, \cdot) r_i = (r_{i,1} K(x_i, \cdot), \dots, r_{i,c} K(x_i, \cdot)). \quad (124)$$

Therefore, using the product structure of $\mathcal{H} = \mathcal{H}_K^c$,

$$\langle K(x_i, \cdot) r_i, v \rangle_{\mathcal{H}} = \sum_{j=1}^c r_{i,j} \langle K(x_i, \cdot), v_j \rangle_{\mathcal{H}_K} = \sum_{j=1}^c r_{i,j} v_j(x_i) = \langle r_i, v(x_i) \rangle_{\mathbb{R}^c}. \quad (125)$$

Hence the Riesz representative of $D\mathcal{L}_{\mathcal{B}}(f; \cdot)$ in $\mathcal{H}$ is

$$\nabla_{\mathcal{H}} \mathcal{L}_{\mathcal{B}}(f)(x) = \frac{1}{m} \sum_{i=1}^m K(x_i, x) (f(x_i) - y_i). \quad (126)$$

F 1-smoothness with respect to the Banach batch norm

Let f and f' be two functions restricted to the batch $\mathcal{B} = \{x_1, \dots, x_m\}$. By the definition of the Banach batch norm,

$$\|f(x_i) - f'(x_i)\|_2 \leq \|f - f'\|_{\mathcal{B}}, \quad i = 1, \dots, m. \quad (127)$$

Therefore,

$$\sum_{i=1}^m \|f(x_i) - f'(x_i)\|_2^2 \leq \sum_{i=1}^m \|f - f'\|_{\mathcal{B}}^2 = m \|f - f'\|_{\mathcal{B}}^2. \quad (128)$$

Multiplying by $1/(2m)$ gives

$$\frac{1}{2m} \sum_{i=1}^m \|f(x_i) - f'(x_i)\|_2^2 \leq \frac{1}{2} \|f - f'\|_{\mathcal{B}}^2. \quad (129)$$

Using this bound in the exact expansion of the batch loss,

$$\mathcal{L}_{\mathcal{B}}(f) = \mathcal{L}_{\mathcal{B}}(f') + D\mathcal{L}_{\mathcal{B}}(f'; f - f') + \frac{1}{2} \|f - f'\|_{L^2(P_{\mathcal{B}})}^2, \quad (130)$$

we obtain

$$\mathcal{L}_{\mathcal{B}}(f) \leq \mathcal{L}_{\mathcal{B}}(f') + D\mathcal{L}_{\mathcal{B}}(f'; f - f') + \frac{1}{2} \|f - f'\|_{\mathcal{B}}^2. \quad (131)$$

Thus, the batch loss is 1-smooth with respect to the Banach batch norm $\|\cdot\|_{\mathcal{B}}$.

G RKHS Gradient Compatibility

Let

$$C_{\mathcal{B}} = F_{\theta, \mathcal{B}} - Y_{\mathcal{B}} = (c_1, \dots, c_m), \quad c_i = f_{\theta}(x_i) - y_i. \quad (132)$$

Evaluating the RKHS gradient in (126) on the batch gives

$$G_{\mathcal{B}} = (\nabla_{\mathcal{H}} \mathcal{L}_{\mathcal{B}}(f_{\theta})(x_1), \dots, \nabla_{\mathcal{H}} \mathcal{L}_{\mathcal{B}}(f_{\theta})(x_m)) = \frac{1}{m} K_{\mathcal{B}} C_{\mathcal{B}}. \quad (133)$$

If the Gram matrix satisfies $K_{\mathcal{B}} \succ 0$, then

$$C_{\mathcal{B}} = m K_{\mathcal{B}}^{-1} G_{\mathcal{B}}. \quad (134)$$

Therefore, using $\|K_{\mathcal{B}}^{-1}\|_2 = 1/\lambda_{\mathcal{B}}$, where $\lambda_{\mathcal{B}} = \lambda_{\min}(K_{\mathcal{B}})$, we obtain

$$\|C_{\mathcal{B}}\|_F \leq \frac{m}{\lambda_{\mathcal{B}}} \|G_{\mathcal{B}}\|_F. \quad (135)$$

Moreover, by Cauchy–Schwarz,

$$\sum_{i=1}^m \|c_i\|_2 \leq \sqrt{m} \|C_{\mathcal{B}}\|_F. \quad (136)$$

Since the Banach batch norm is the maximum pointwise norm,

$$\|G_{\mathcal{B}}\|_F \leq \sqrt{m} \|\nabla_{\mathcal{H}} \mathcal{L}_{\mathcal{B}}(f_{\theta})\|_{\mathcal{B}}. \quad (137)$$

Combining (135), (136), and (137) gives

$$\frac{1}{m} \sum_{i=1}^m \|c_i\|_2 \leq \frac{m}{\lambda_{\mathcal{B}}} \|\nabla_{\mathcal{H}} \mathcal{L}_{\mathcal{B}}(f_{\theta})\|_{\mathcal{B}}. \quad (138)$$

For any h such that $\|h\|_{\mathcal{B}} \leq 1$,

$$|D\mathcal{L}_{\mathcal{B}}(f_{\theta}; h)| \leq \frac{1}{m} \sum_{i=1}^m \|c_i\|_2. \quad (139)$$

Taking the supremum over all such h and using (138), we obtain

$$\|D\mathcal{L}_{\mathcal{B}}(f_{\theta}; \cdot)\|_{\mathcal{B},*} \leq \beta_{\mathcal{K}} \|\nabla_{\mathcal{H}} \mathcal{L}_{\mathcal{B}}(f_{\theta})\|_{\mathcal{B}}, \quad \beta_{\mathcal{K}} = \frac{m}{\lambda_{\mathcal{B}}}. \quad (140)$$

H RKHS Descent Constant

With the notation of Appendix G, the directional derivative along the RKHS gradient is

$$D\mathcal{L}_{\mathcal{B}}(f_{\theta}; \nabla_{\mathcal{H}} \mathcal{L}_{\mathcal{B}}(f_{\theta})) = \frac{1}{m} \sum_{i=1}^m \langle c_i, \nabla_{\mathcal{H}} \mathcal{L}_{\mathcal{B}}(f_{\theta})(x_i) \rangle_{\mathbb{R}^c}. \quad (141)$$

Using $G_{\mathcal{B}} = (1/m)K_{\mathcal{B}}C_{\mathcal{B}}$, this can be written as

$$D\mathcal{L}_{\mathcal{B}}(f_{\theta}; \nabla_{\mathcal{H}} \mathcal{L}_{\mathcal{B}}(f_{\theta})) = \frac{1}{m^2} \text{tr}(C_{\mathcal{B}}^{\top} K_{\mathcal{B}} C_{\mathcal{B}}). \quad (142)$$

Since $K_{\mathcal{B}} \succeq \lambda_{\mathcal{B}} I_m$, it follows that

$$\text{tr}(C_{\mathcal{B}}^{\top} K_{\mathcal{B}} C_{\mathcal{B}}) \geq \lambda_{\mathcal{B}} \|C_{\mathcal{B}}\|_F^2. \quad (143)$$

For the normalized RBF kernel, $|K_{\sigma}(x_i, x_j)| \leq 1$. Therefore,

$$\|\nabla_{\mathcal{H}} \mathcal{L}_{\mathcal{B}}(f_{\theta})\|_{\mathcal{B}} \leq \frac{1}{m} \sum_{i=1}^m \|c_i\|_2 \leq \frac{1}{\sqrt{m}} \|C_{\mathcal{B}}\|_F. \quad (144)$$

Thus,

$$\|C_{\mathcal{B}}\|_F^2 \geq m \|\nabla_{\mathcal{H}} \mathcal{L}_{\mathcal{B}}(f_{\theta})\|_{\mathcal{B}}^2. \quad (145)$$

Combining (142), (143), and (145), we obtain

$$D\mathcal{L}_{\mathcal{B}}(f_{\theta}; \nabla_{\mathcal{H}} \mathcal{L}_{\mathcal{B}}(f_{\theta})) \geq \frac{\lambda_{\mathcal{B}}}{m} \|\nabla_{\mathcal{H}} \mathcal{L}_{\mathcal{B}}(f_{\theta})\|_{\mathcal{B}}^2. \quad (146)$$

Hence the RKHS descent constant is

$$\alpha_{\mathcal{K}} = \frac{\lambda_{\mathcal{B}}}{m}. \quad (147)$$

I RKHS PL condition in the Banach batch norm

For the RKHS formulation evaluated on the batch, we verify a Polyak–Łojasiewicz inequality of the form

$$\mathcal{L}_{\mathcal{B}}(f) - \mathcal{L}_{\mathcal{B}}^* \leq \frac{1}{2\mu_{\mathcal{K}}} \|D\mathcal{L}_{\mathcal{B}}(f; \cdot)\|_{\mathcal{B},*}^2. \quad (148)$$

For the empirical quadratic loss, if the batch is interpolable, then $\mathcal{L}_{\mathcal{B}}^* = 0$. From (108),

$$\mathcal{L}_{\mathcal{B}}(f) = \frac{1}{2m} \|C_{\mathcal{B}}\|_F^2. \quad (149)$$

Moreover, by the definition of the dual norm induced by (106),

$$\|D\mathcal{L}_{\mathcal{B}}(f; \cdot)\|_{\mathcal{B},*} = \frac{1}{m} \sum_{i=1}^m \|c_i\|_2. \quad (150)$$

Therefore,

$$\|D\mathcal{L}_{\mathcal{B}}(f; \cdot)\|_{\mathcal{B},*}^2 = \frac{1}{m^2} \left(\sum_{i=1}^m \|c_i\|_2 \right)^2. \quad (151)$$

Since

$$\left(\sum_{i=1}^m \|c_i\|_2 \right)^2 \geq \sum_{i=1}^m \|c_i\|_2^2 = \|C_{\mathcal{B}}\|_F^2, \quad (152)$$

we obtain

$$\|D\mathcal{L}_{\mathcal{B}}(f; \cdot)\|_{\mathcal{B},*}^2 \geq \frac{1}{m^2} \|C_{\mathcal{B}}\|_F^2. \quad (153)$$

Equivalently,

$$\|C_{\mathcal{B}}\|_F^2 \leq m^2 \|D\mathcal{L}_{\mathcal{B}}(f; \cdot)\|_{\mathcal{B},*}^2. \quad (154)$$

Substituting this bound into (149) gives

$$\mathcal{L}_{\mathcal{B}}(f) \leq \frac{m}{2} \|D\mathcal{L}_{\mathcal{B}}(f; \cdot)\|_{\mathcal{B},*}^2. \quad (155)$$

Comparing (155) with (148) yields

$$\frac{1}{2\mu_{\mathcal{K}}} = \frac{m}{2}, \quad \mu_{\mathcal{K}} = \frac{1}{m}. \quad (156)$$